Image and Voice Pipelines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

TEACHER GUIDE · 1 OF 9

Image and Voice Pipelines

> Prepare Image and Voice Pipelines: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of speech **pipeline**s and multimodal **evidence**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/26, ai-3/12, ai-3/13. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Text as Pieces

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

Look for: Encode and decode text under an explicit toy vocabulary

Reference example, not the only acceptable wording: Teacher explanation: The rule chooses the longest matching ordinary token; sat has length three and s length one.

Vocabulary and matching are case sensitive and contain no uppercase C.

Recommendation Scores

Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a weighted score

Reference example, not the only acceptable wording: Score 8.

Repair Bias in Data

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

Look for: Identify selection and coverage bias

Reference example, not the only acceptable wording: Afternoon or evening visitors, and people unable to attend in the morning.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A voice request says 'under 90 pages'. The transcript says 'under 19 pages'; filtering correctly finds no books. Identify the first failing stage and a useful confirmation step.

Concept to explain

Perception systems transform **signal**s through multiple fallible stages.

Term	Meaning
signal	A measurable quantity carrying information, such as changing sound pressure or light intensity.
pipeline	A connected sequence of stages in which one stage's output becomes another's input.
recognition	Interpreting an input as a known category or pattern rather than merely recording its signal.
stage error	A failure introduced at a particular step of a pipeline.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. A multimodal **pipeline** may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can trace where speech information changes and resolve a declared-authority conflict.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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Explain the mechanism

> Model Image and Voice Pipelines using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Voice: sound → sampled **signal** → recognized words → intent → action.

2. Image: pixels → features → score → label.

3. Noisy sound can produce wrong words even if the intent stage works.

4. Diagnose the first wrong stage before changing later stages.

Recalculate, retrace, or restate the reasoning in your own representation:

> CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

> ### The teaching case
>
> A voice book helper transcribes speech, extracts preference, then ranks books.

> Where can a wrong suggestion begin?

Exact explanation to model

> ### Why the result follows
>
> **Transcription** can mishear poetry and pass the wrong preference downstream. Check intermediate results.

Explain the added worked example

A multimodal **pipeline** may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Explain the added worked example · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately. Compare conflicting **evidence** without assuming one **modality** always wins.

Ask for clarification when the task needs unresolved information.

Explain the added worked example · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Modality	A form of information such as audio, image or text
Transcription	A text representation of speech
Error propagation	An earlier error affecting later processing

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Model the reasoning aloud · 3

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Image and Voice Pipelines

A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup.

The facts we will use

Supplied facts

Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

Supplied facts

A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.

AI ACADEMY / AI-2/26

Image and Voice Pipelines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Audio input

Known phrase names fourteen, not forty.



2. Transcription

Text output changes the shelf number to forty.



3. Cross-check

The authoritative inventory gives shelf 14.



4. Review and correct

Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

CHANGE ONE THING

Remove access to the audio, while keeping the authoritative inventory.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Audio input	Known phrase names fourteen, not forty.
2. Transcription	Text output changes the shelf number to forty.
3. Cross-check	The authoritative inventory gives shelf 14.
4. Review and correct	Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A correct later lookup cannot repair an earlier wrong **input** by magic. Locate the first difference between known audio and transcript, then compare the stated source authority.

Keeping both records makes the correction auditable. A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

What this example does not establish

Keep the stated limit

Declared authority resolves this task's source choice; it does not make every inventory entry universally accurate.

Change one thing in the pictured case

Change one thing

Change: Remove access to the audio, while keeping the authoritative inventory.

Predicted result

Prediction: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked.

Why the result follows

Reason: The inventory resolves the shelf task; it does not recreate unavailable audio.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Image and Voice Pipelines, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A voice book helper transcribes speech, extracts preference, then ranks books.

2. State the decision to explain

Write the question to explain: Where can a wrong suggestion begin?

3. Reveal the worked reasoning

Reveal after **prediction**s: **Transcription** can mishear poetry and pass the wrong preference downstream. Check intermediate results.

Hold back the independent task

Keep for an independent attempt: A voice helper hears find a book, converts speech to text, then searches. It writes find a cook. Which stage first failed and what should you inspect?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

Stage	Teacher action
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion practice

Completion task: Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Speech **pipeline**s and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which stage first fails, and why should the faulty transcript not be silently overwritten?

Reasoned teacher answer

Transcription first fails. Preserve it to diagnose the confusion and distinguish correction from the original model **output**.

Changed-case reasoning: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked. The inventory resolves the shelf task; it does not recreate unavailable audio.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Image and Voice Pipelines investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Trace four failures backward through stage cards. Change one stage **input**, predict downstream effects, and propose a stage-specific test.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome

Record	What to capture
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Image and Voice Pipelines questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Transcription can mishear poetry and pass the wrong preference downstream. Check intermediate results.

Independent question

Independent question · withhold the key

A voice helper hears find a book, converts speech to text, then searches. It writes find a cook. Which stage first failed and what should you inspect?

Expected independent answer

Teacher key · Expected independent answer

The speech-to-text stage first introduced the wrong word. Check recording clarity and

transcription before blaming the search results.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “An incorrect final answer means every stage failed.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

The **transcription** stage altered the number. Confirm 'under 90 pages' with the user or inspect the transcript before filtering.

Fixing the ranking stage cannot restore a lost constraint; keep intermediate records for diagnosis without unnecessary personal recordings.

Explanations for the additional workbook examples

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: A multimodal **pipeline** may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately. Compare conflicting **evidence** without assuming one **modality** always wins.

Ask for clarification when the task needs unresolved information.

W2 Trace the worked case

Workbook question · W2

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

Teacher reasoning · W4

Teacher explanation: Report F from the authoritative approval, noting the transcript conflict if relevant. This resolves the task authority, not what the original speaker actually said.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

The new case should test this particular **mechanism** rather than merely restate the original result.

Assess independent transfer

Independent question · withhold the key

Selected case: Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

Selected case answer

> **Teacher key · Selected case answer**
>
> Teacher explanation: Report F from the authoritative approval, noting the transcript conflict if relevant. This resolves the task authority, not what the original speaker actually said.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Image and Voice Pipelines to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

If voice or image **input** is added, show each transformation and a safe fallback.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 27: Predictive vs Generative. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Report F from the authoritative approval, noting the transcript conflict if relevant. This resolves the task authority, not what the original speaker actually said. Require intermediate reasoning and the stated scope.

Fresh retry: Transcript says eight; a validated order record says eighteen and is declared authoritative for ordering.

Expected reasoning: Use eighteen for the bounded order-record task; retain the transcript discrepancy and do not claim perfect **transcription**.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Retrieval and indexing. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-2/27, ai-2/29, ai-2/32. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Image and Voice Pipelines lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of speech **pipeline**s and multimodal **evidence**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Check every learner before discussion: Use the worked case for Speech **pipeline**s and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the

reasoning goal.

Extend the reasoning

Extension: Remove access to the audio, while keeping the authoritative inventory. Require a **prediction** and a reason before comparing.

Expected result: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked. Reason: The inventory resolves the shelf task; it does not recreate unavailable audio.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

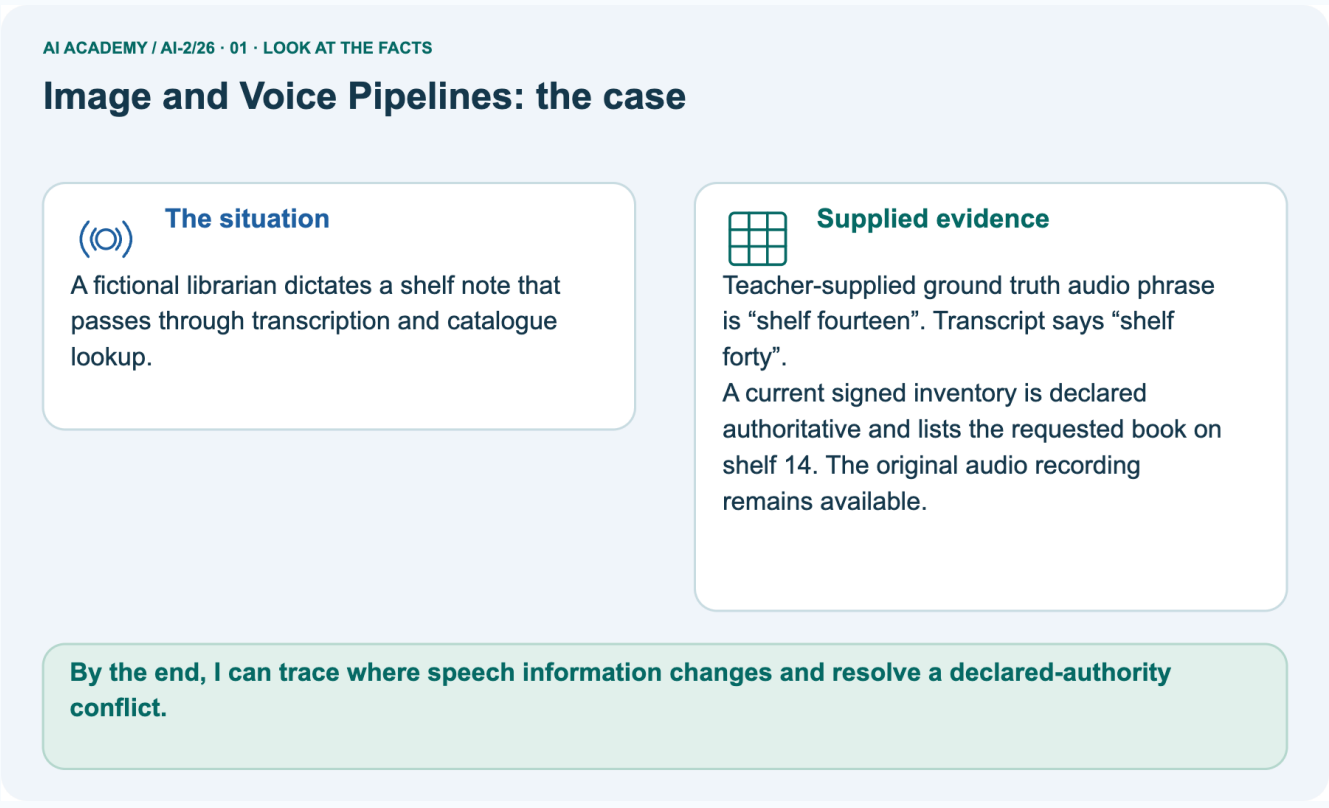
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Image and Voice Pipelines graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Image and Voice Pipelines: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional librarian dictates a shelf note that passes through **transcription** and catalogue lookup. Supplied facts: Teacher-supplied ground truth audio phrase is “shelf fourteen”. Transcript says “shelf forty”.

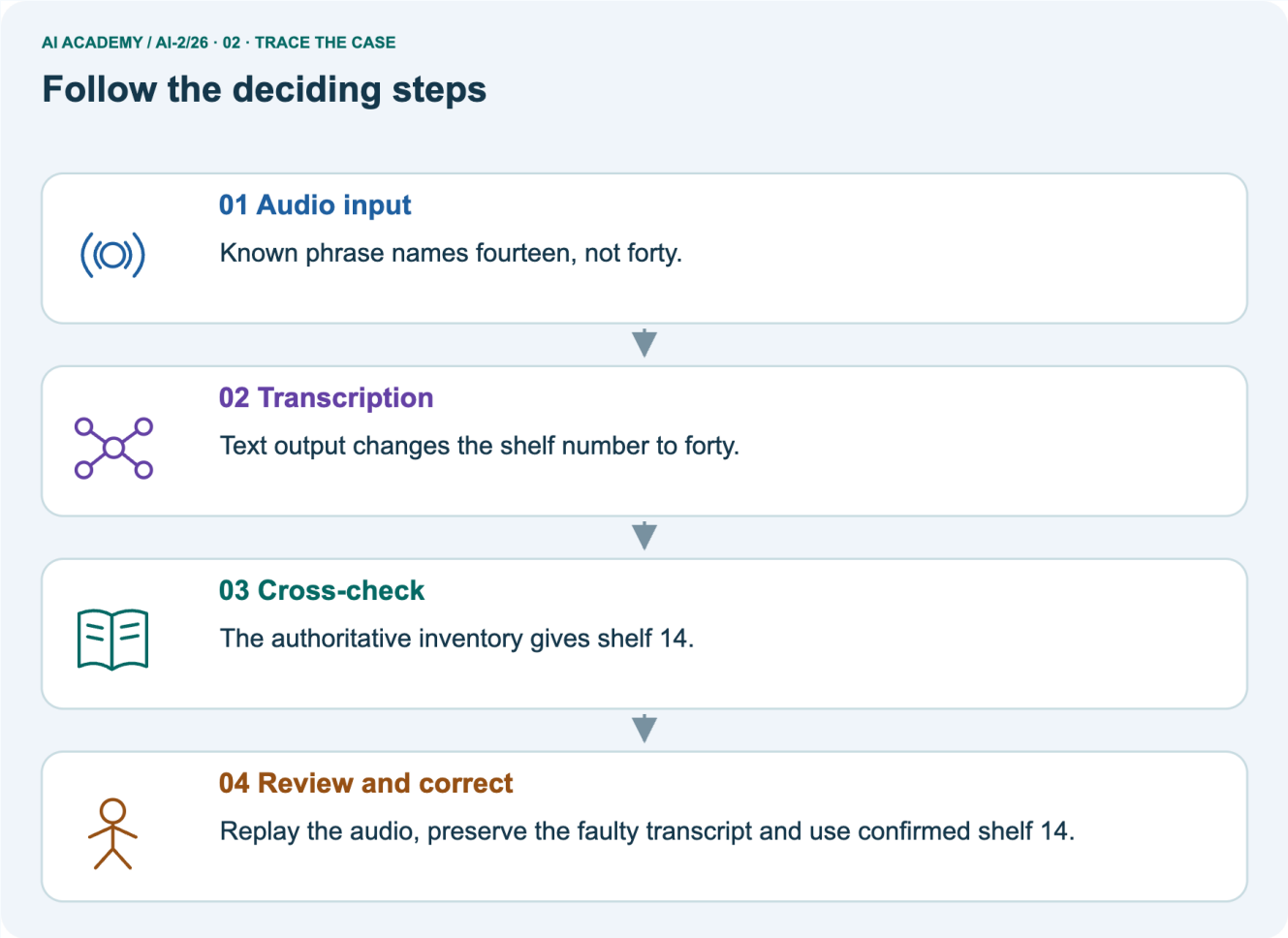
A current signed inventory is declared authoritative and lists the requested book on shelf 14. The original audio recording remains available.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

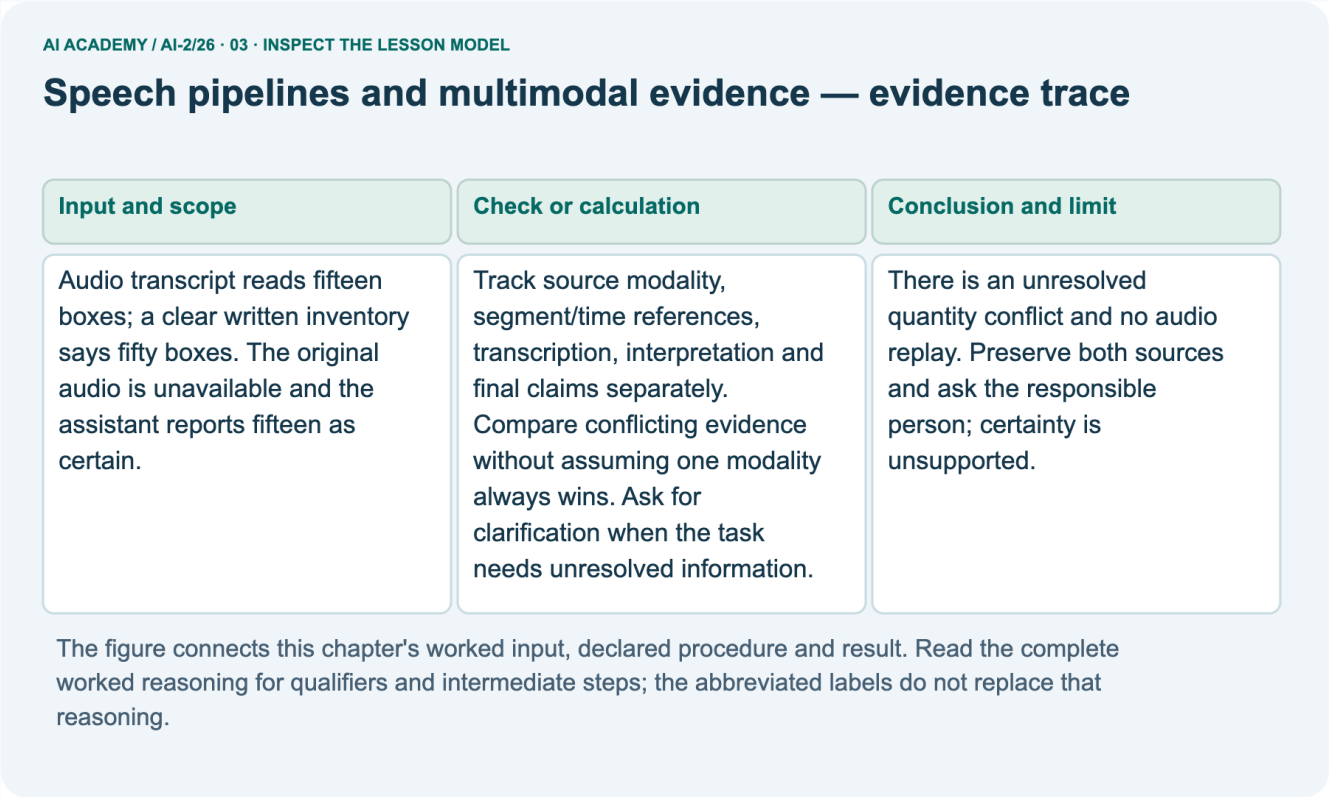
Audio **input**: Known phrase names fourteen, not forty. → **Transcription**: Text **output** changes the shelf number to forty. → Cross-check: The authoritative inventory gives shelf 14. → Review and correct: Replay the audio, preserve the faulty transcript and use confirmed shelf 14.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech **pipeline**s and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

Compare conflicting **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity conflict and no audio replay.

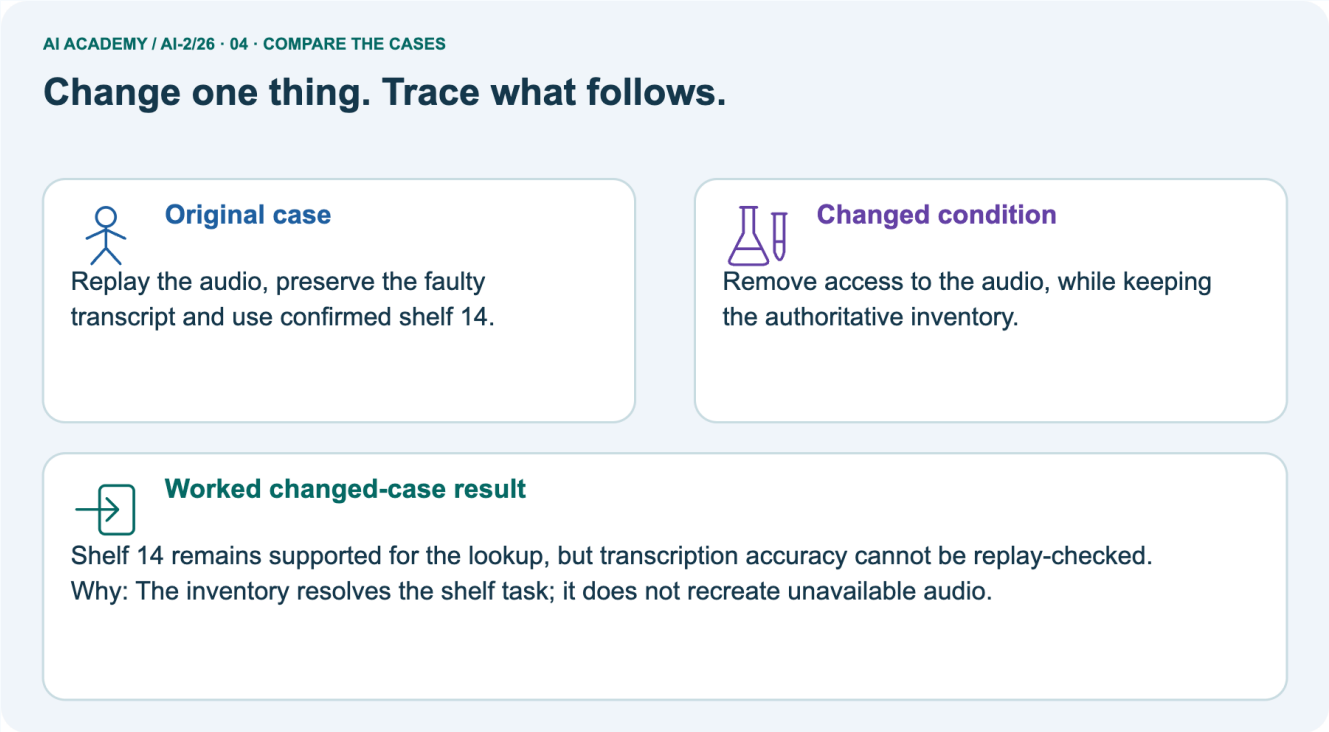
Preserve both sources and ask the responsible person; certainty is unsupported.

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Replay the audio, preserve the faulty transcript and use confirmed shelf 14. Changed condition: Remove access to the audio, while keeping the authoritative inventory.

Expected result: Shelf 14 remains supported for the lookup, but **transcription** accuracy cannot be replay-checked. Reason: The inventory resolves the shelf task; it does not recreate unavailable audio.

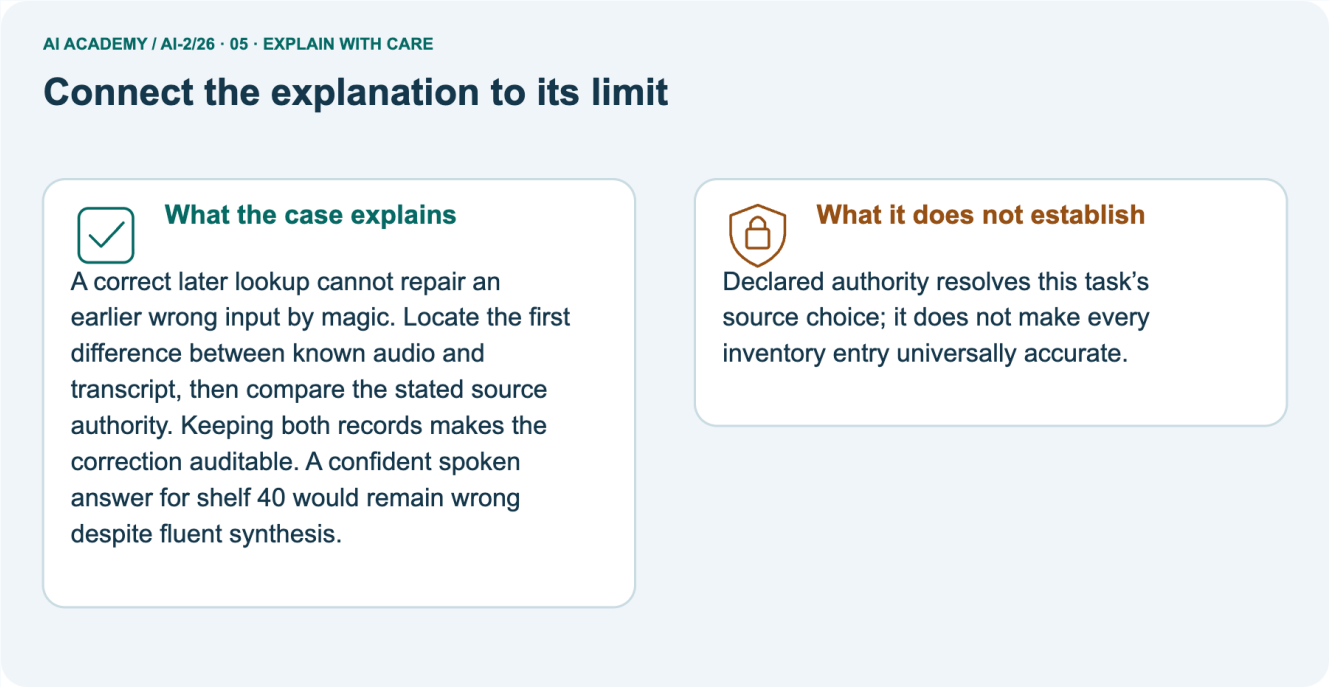
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The inventory resolves the shelf task; it does not recreate unavailable audio.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A correct later lookup cannot repair an earlier wrong **input** by magic. Locate the first difference between known audio and transcript, then compare the stated source authority.

Keeping both records makes the correction auditable. A confident spoken answer for shelf 40 would remain wrong despite fluent synthesis.

Limit: Declared authority resolves this task’s source choice; it does not make every inventory entry universally accurate.

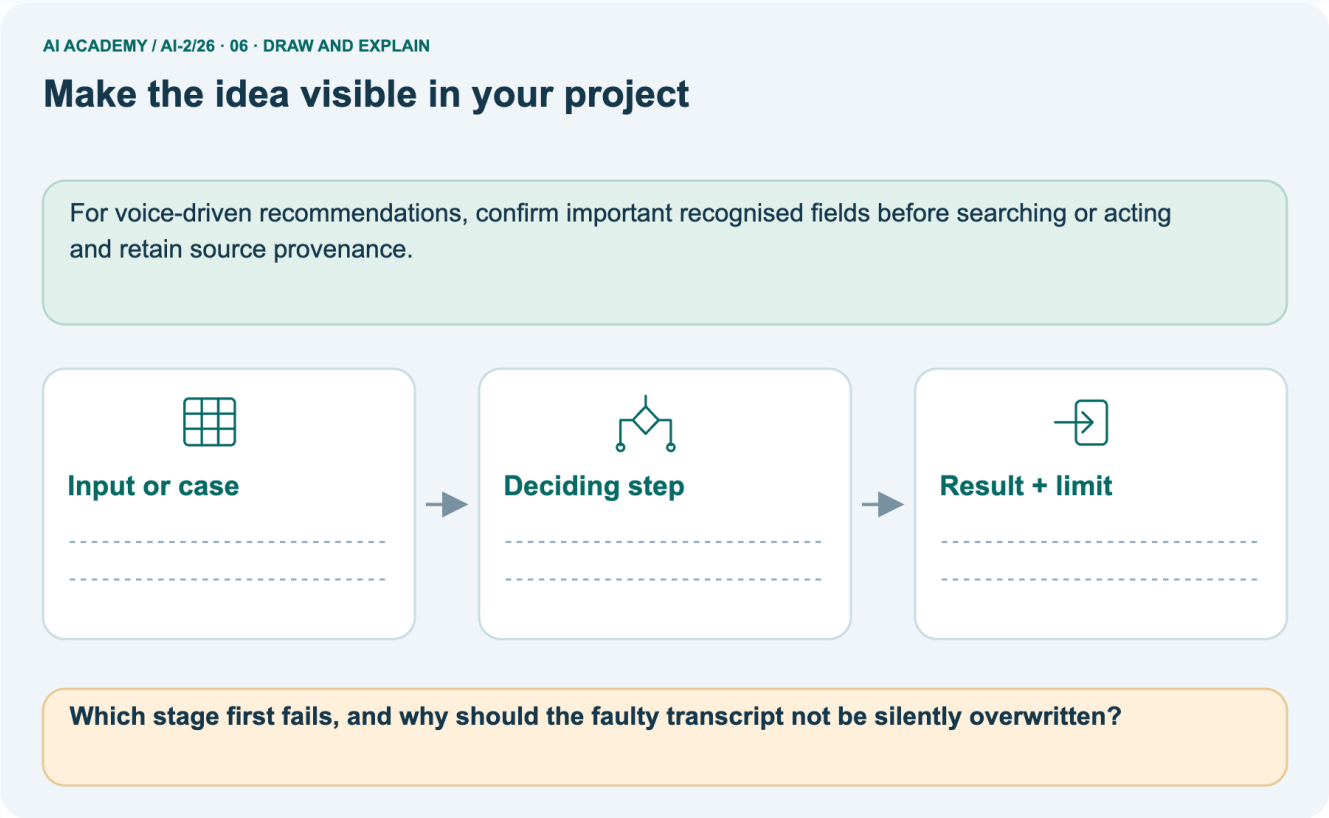
> Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Declared authority resolves this task’s source choice; it does not make every inventory entry universally accurate.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For voice-driven recommendations, confirm important recognised fields before searching or acting and retain source provenance. Draw your **input**, deciding step, result and limit.

Supported question: Which stage first fails, and why should the faulty transcript not be silently overwritten?

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: **Transcription** first fails. Preserve it to diagnose the confusion and distinguish correction from the original model **output**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.